

Machine Learning Analysis Report

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Overview

This project addresses a supervised binary classification problem using the Breast Cancer Wisconsin Diagnostic dataset from UCI: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>. I built a scikit-learn pipeline with standard scaling and logistic regression to classify observations as benign or malignant. The goal is to demonstrate model design, evaluation, and responsible interpretation, not clinical deployment.

Dataset Description

The dataset contains 569 rows and 32 columns in the saved project table, including the target and readable diagnosis label. Each row describes computed features from a digitized breast mass image, including radius, texture, perimeter, area, smoothness, compactness, concavity, and related summaries. The target variable identifies malignant or benign diagnosis. This dataset is a widely used benchmark, but its controlled setting means performance should not be treated as proof of real-world clinical reliability.

Modeling Approach

The notebook uses a stratified train-test split so both classes are represented in training and evaluation. Logistic regression was selected because it is a strong, interpretable baseline for tabular binary classification and works well with standardized continuous predictors. Accuracy, class precision, class recall, confusion matrix, and ROC AUC were appropriate because the task has asymmetric risk: false benign predictions may be more serious than false malignant predictions. The workflow uses scikit-learn pipeline conventions consistent with the library design described by Pedregosa et al. (2011).

Results

The executed notebook produced test accuracy of 0.986 and ROC AUC of 0.998. Malignant precision was 0.981, and malignant recall was 0.981. The confusion matrix was $\begin{bmatrix} 52 & 1 \\ 1 & 89 \end{bmatrix}$. Figure 1 shows the confusion matrix, Figure 2 shows the ROC curve, and Figure 3 identifies influential coefficients such as worst texture, radius error, worst symmetry, worst concave points, area error.

Interpretation for a Non-Technical Audience

The model separated the two classes well on the held-out benchmark test set. In practical terms, it learned numeric measurement patterns that are strongly associated with the diagnostic label. However, a high score on a benchmark dataset does not mean the model is ready to guide patient care. A real system would need external validation, clinical review, calibration, monitoring, and a clear human-in-the-loop process.

Limitations and Potential Bias

The project depends on a curated benchmark dataset and does not include all demographic, equipment, site, or workflow variation that may appear in production. Bias could arise if the original sample does not represent the population where the model is used. The model is also linear, so it may miss nonlinear interactions; the tradeoff is improved transparency and a simpler baseline for explanation.

References

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
<https://jmlr.org/papers/v12/pedregosa11a.html>

Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5DW2B>

NIST. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). <https://doi.org/10.6028/NIST.AI.100-1>

Selected Figures

Figure 1. Confusion matrix.

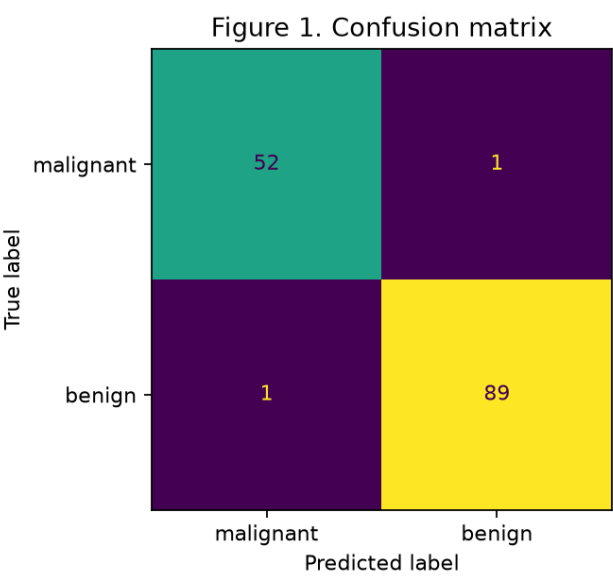


Figure 2. ROC curve.

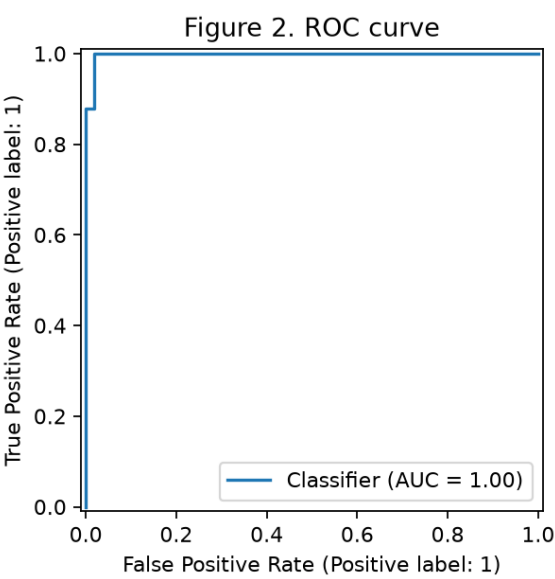


Figure 3. Largest logistic regression coefficients.

Figure 3. Largest absolute logistic regression coefficients

